

Quantum coherent beating in polar disubphthalocyanine-fullerene cocrystals for ultrafast photothermal conversion

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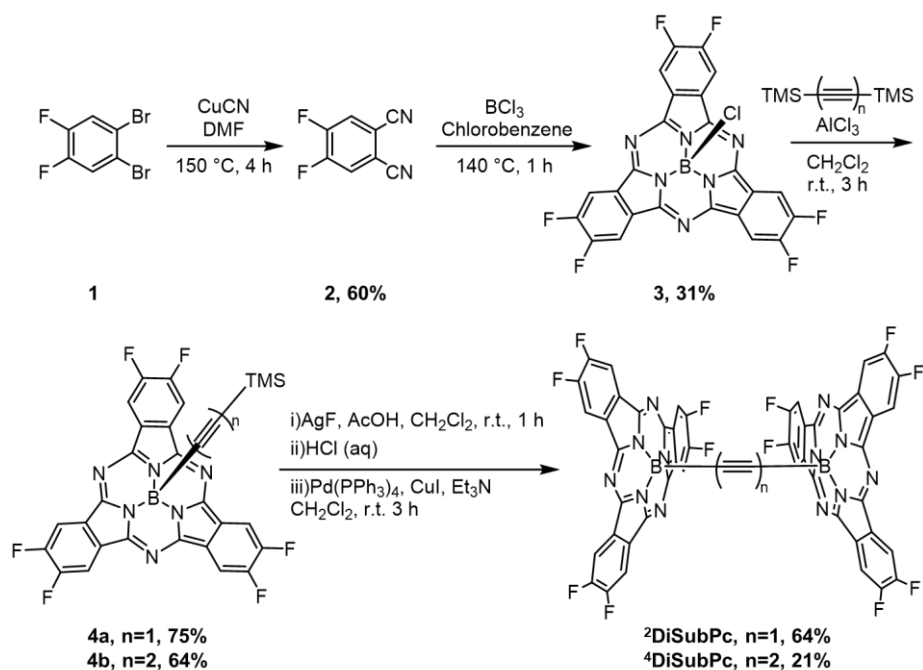
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I. General information

All commercially available reagents were used without further purification unless otherwise noted. All the reactions dealing with air- or moisture-sensitive compounds were carried out in a dry reaction vessel under a positive pressure of nitrogen.

NMR spectra were recorded on an Agilent 400-MR DD2 spectrometer. The ^1H NMR (400 MHz) chemical shifts were measured relative to CDCl_3 (CDCl_3 : $\delta = 7.26$ ppm). The ^{13}C NMR (100 MHz) chemical shifts were given using CDCl_3 as the internal standard (CDCl_3 : $\delta = 77.16$ ppm). Matrix-assisted laser desorption/ionization time of flight mass spectra (MALDI-TOF-MS) were collected with a Shimadzu AXIMA Performance. UV-vis-NIR absorption spectra were recorded by SHIMADZU UV-3600. Emission spectra were obtained using an Edinburgh Instruments FLS920 Fluorimeter. PL lifetimes were acquired using the Fluorimeter pulsed xenon flash-lamp, Xe900, of 0.4 W and Picoquant PLS- λ nanosecond LED diodes. X-ray single-crystal diffraction data were collected on an Agilent Technologies Gemini single-crystal diffractometer. Thermogravimetric analysis (TGA) was carried out using METTLER TOLEDO TGA/DSC2 at a rate of 10 $^\circ\text{C}/\text{min}$ under nitrogen atmosphere. Cyclic voltammogram (CV) was performed on LK2005A with a solution of tetrabutylammonium hexafluorophosphate (Bu_4NPF_6 , 0.1 M) in CH_2Cl_2 as electrolyte and ferrocene/ferrocenium (Fc/Fc^+) as standard. Three-electrode system (Ag/Ag^+ , platinum wire and glassy carbon electrode as reference, counter, and work electrode, respectively) was used in the CV measurement. All potentials were corrected against Fc/Fc^+ . CV was measured with a scan rate of 100 mV/s. Theoretical calculations were carried out using Gaussian 09 software. The FMO distributions were visualized using Gaussview 5.0 software. The ground state structures were optimized by density functional theory (DFT) at B3LYP/6-31G* level.

II. Synthesis

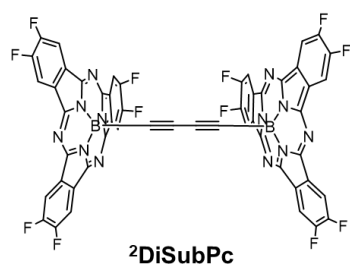


Scheme S1. Synthesis routes of **DiSubPcs**.

Preparation of the 4a and 4b

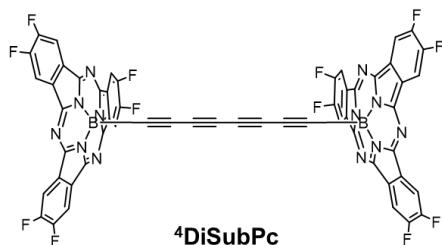
1,2-Dibromo-4,5-difluorobenzene **1** were obtained from commercial suppliers and used without further purification, **2-4** were prepared according to the literature¹. The ¹H NMR and ¹³C NMR data were in accordance with the reported data.

Preparation of the DiSubPcs



4a (600 mg, 1 mmol) was dissolved in a solution of CH₂Cl₂ (50 mL) and AcOH (5 mL, 0.7 M in CH₂Cl₂), and then AgF (1.3 g, 10 mmol) was added. The mixture was allowed to stir for 60 min, and then aqueous HCl (100 mL, 0.1 M) was added. The reaction mixture was vigorously stirred and extracted with CH₂Cl₂, and the combined organic

phases was dried over Na₂SO₄ and filtered. The volume was concentrated to 50 mL, then Pd(PPh₃)₂Cl₂ (72 mg, 0.1 mmol) and CuI (38 mg, 0.2 mmol) was added, and Et₃N (5 mL) was injected to the system. The mixture was stirred for 4 h, filtered through a celite pad and washed with CH₂Cl₂, and concentrated under vacuum. The residue was chromatographed on silica gel with PE/CH₂Cl₂ = 1/1 (v/v) to provide the corresponding product **²DiSubPc** as red solid (64% yield). ¹H NMR (400 MHz, CDCl₃): δ = 8.42 (t, *J* = 8.0 Hz, 2H), ¹³C NMR (100 MHz, CDCl₃): δ = 153.98, 153.81, 151.42, 151.26, 149.40, 127.01, 110.40, 29.85. MS (MALDI-TOF): calcd for C₅₂H₁₂B₂F₁₂N₁₂: 1054.1302, found: 1054.1840.



⁴DiSubPc was synthesized according the same route with the yield of 21%. ¹H NMR (400 MHz, CDCl₃): δ = 8.53 (t, *J* = 8.0 Hz, 2H), ¹³C NMR (100 MHz, CDCl₃): δ = 154.10, 153.94, 151.54, 151.38, 149.63, 127.11, 110.56, 61.10, 60.82, MS (MALDI-TOF): calcd for C₅₆H₁₃B₂F₁₂N₁₂⁺: [M+H]⁺, 1103.1375, found: 1103.0451.

III. Stability

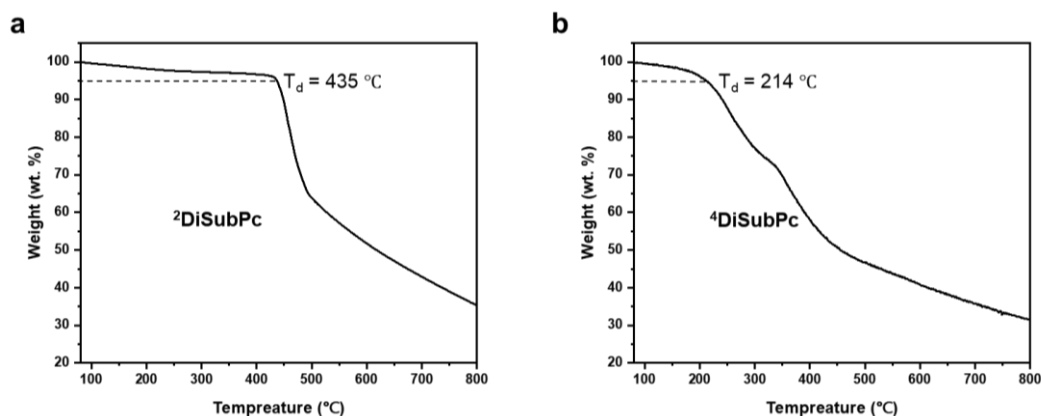


Figure S1. Thermal gravimetric analysis (TGA) curves of compounds (a) $^2\text{DiSubPc}$, (b) $^4\text{DiSubPc}$.

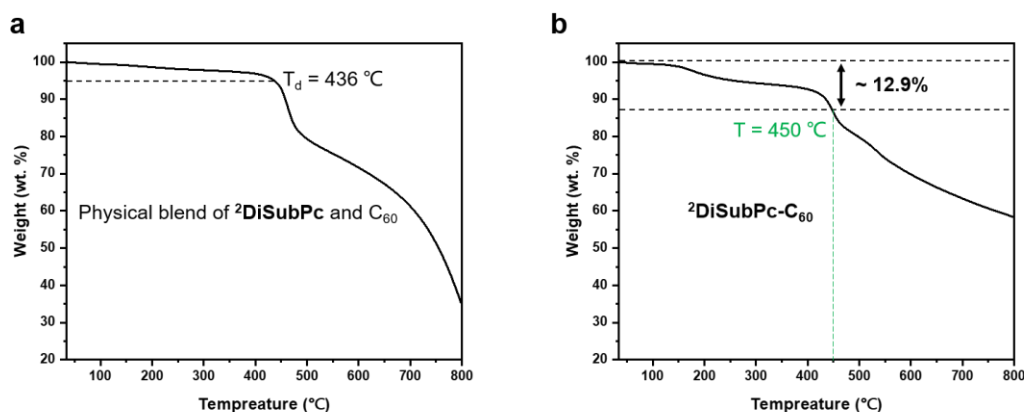


Figure S2. The TGA curves of (a) physical blend of $^2\text{DiSubPc}$ and C_{60} and (b) $^2\text{DiSubPc-C}_{60}$ cocrystals involving toluene.

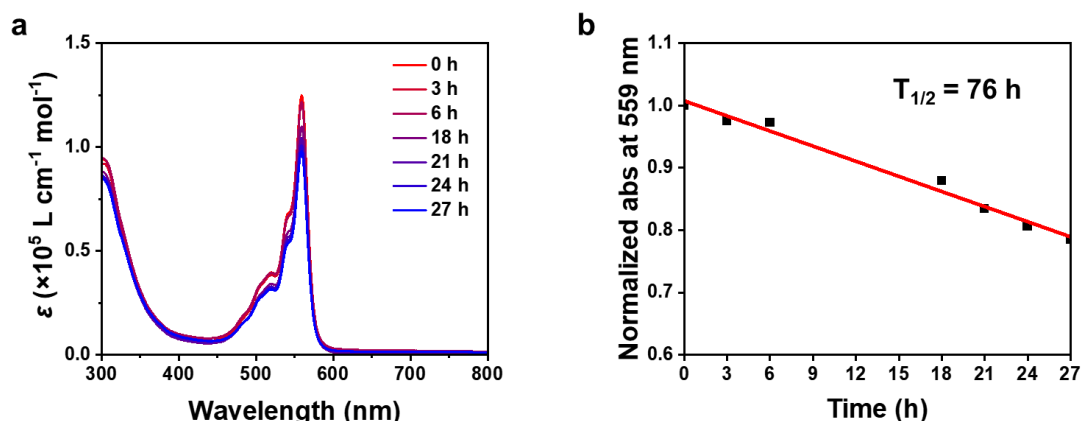


Figure S3. Solution stability evaluation of $^4\text{DiSubPc}$. (a) time-dependent UV-vis absorption spectra of $^4\text{DiSubPc}$ in a 10^{-5} M toluene solution under ambient conditions. (b) Temporal change in absorbance at 559 nm, with an estimated half-life of 76 hours.

IV. Electrochemical and photophysical properties of SubPcs

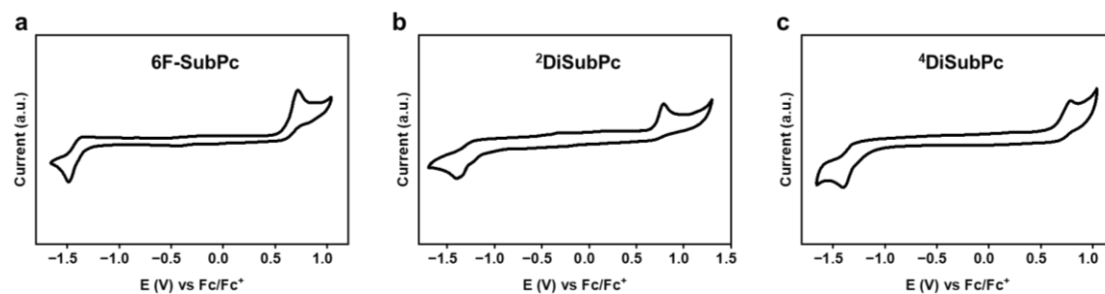


Figure S4. CV of (a) **6F-SubPc**, (b) **²DiSubPc** and (c) **⁴DiSubPc** in dry CH₂Cl₂ (1.0×10⁻³ M) containing 0.1 M Bu₄NPF₆.

Table S1. Photophysical and electrochemical properties of SubPcs through UV/Vis/NIR absorption spectra in toluene and CVs.

Compounds	$\lambda_{\max}^{[a]}$ [nm]	$E_g^{\text{abs [b]}}$ [eV]	HOMO ^[c] [eV]	LUMO ^[c] [eV]	$E_g^{\text{CV[d]}}$ [eV]
6F-SubPc	563	2.25	-5.70	-2.93	2.77
²DiSubPc	569	2.25	-5.65	-2.93	2.67
⁴DiSubPc	574	2.25	-5.70	-2.98	2.72

[a] Absorption maxima in toluene (10 μ M). [b] Optical bandgaps were calculated according to $E_g = 1240/\lambda_{\text{onset}}$, where λ_{onset} is the onset value of the absorption spectrum in the long wavelength region. [c] From CVs measured in CH₂Cl₂ (1.0×10⁻³ M). The HOMO and LUMO energy levels are adjusted according to the redox half potential of Fc/Fc⁺ and estimated according to the formula: E_{HOMO} (eV) = - (4.8 + $E^{\text{ox}} - E_{(\text{Fc}/\text{Fc}^+)^{1/2}}$), E_{LUMO} (eV) = - (4.8 + $E^{\text{red}} - E_{(\text{Fc}/\text{Fc}^+)^{1/2}}$). [d] Estimated according to the formula: E_g (eV) = $E_{\text{LUMO}} - E_{\text{HOMO}}$.

Absorption titrations

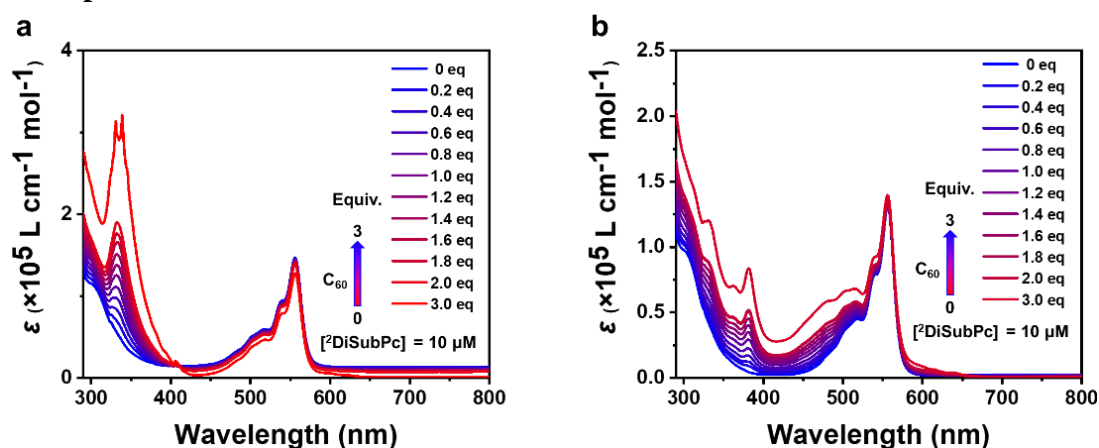


Figure S5. The changes of the UV-Vis-NIR spectra of $^2\text{DiSubPc}$ (10 μM in toluene) with the addition of (a) C_{60} and (b) C_{70} from 0 to 3.0 equiv.

Binding constant based on fluorescence titrations between DiSubPcs and both C_{60} and C_{70}

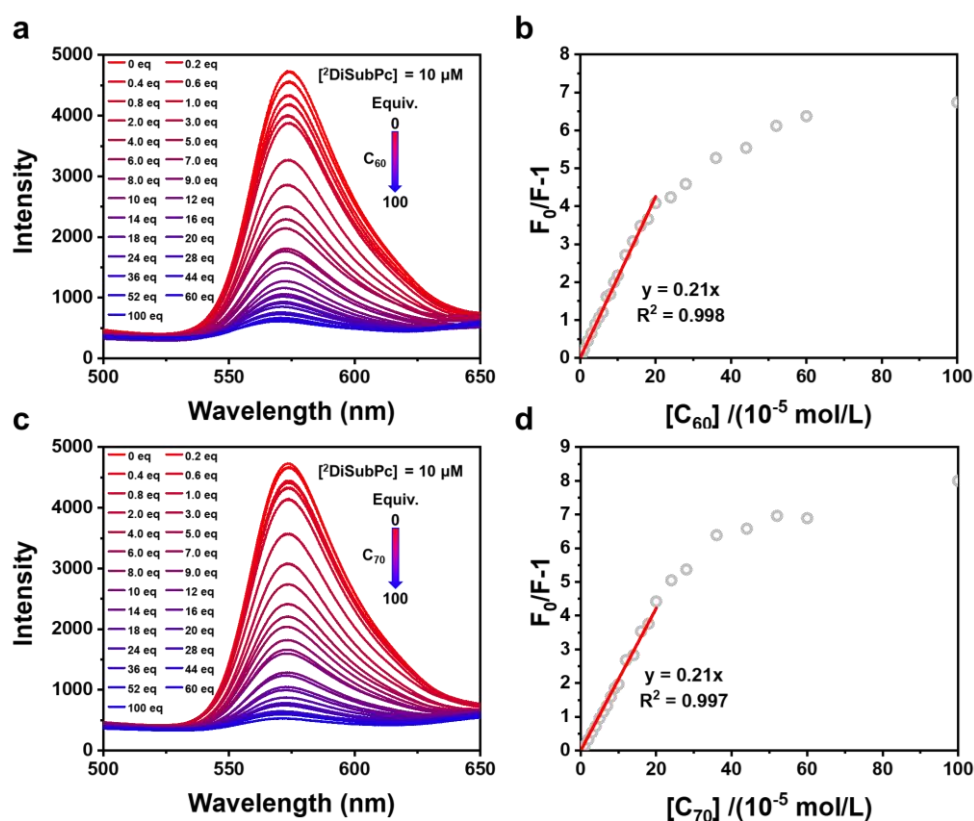


Figure S6. Changes in the PL intensity of $^2\text{DiSubPc}$ (10 μM in toluene) upon the addition of (a) C_{60} and (c) C_{70} from 0 to 100 equiv. (b, d) The corresponding intensity changes of PL at 573 nm (black dots) employed for linear fitting, with the fitting curve displayed in red (0-20 equiv.).

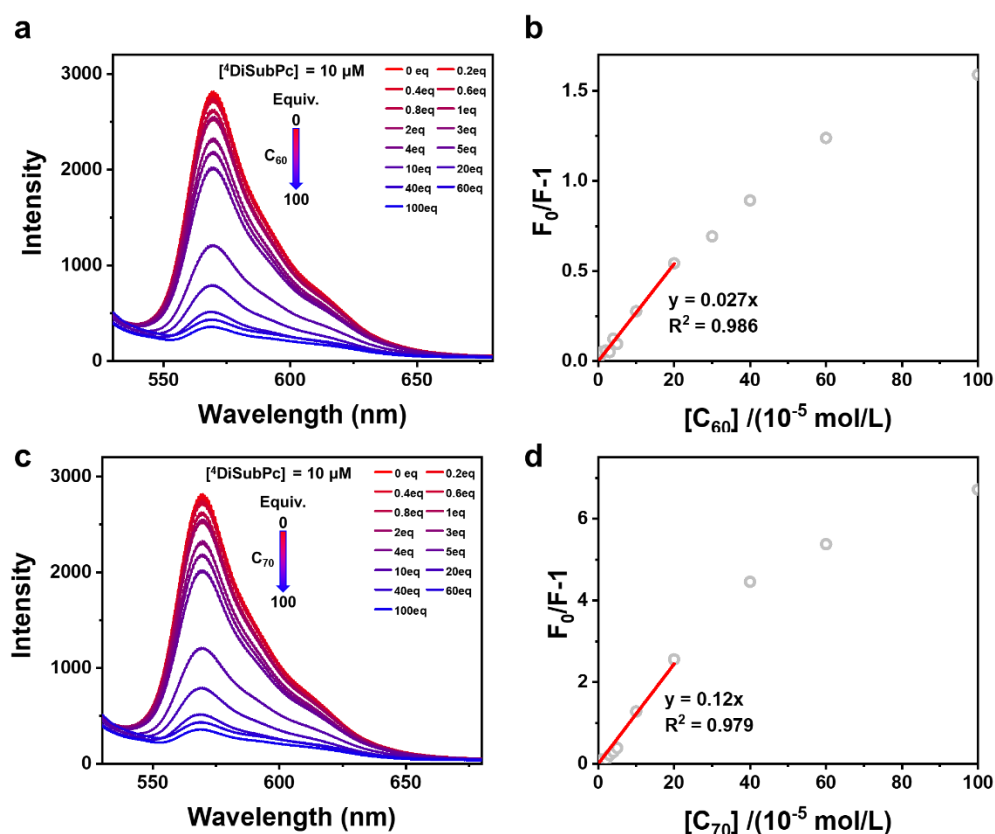


Figure S7. Changes in the PL intensity of $^4\text{DiSubPc}$ (10 μM in toluene) upon the addition of (a) C_{60} and (c) C_{70} from 0 to 100 equiv. (b, d) The corresponding intensity changes of PL at 573 nm (black dots) employed for linear fitting, with the fitting curve displayed in red (0-20 equiv.).

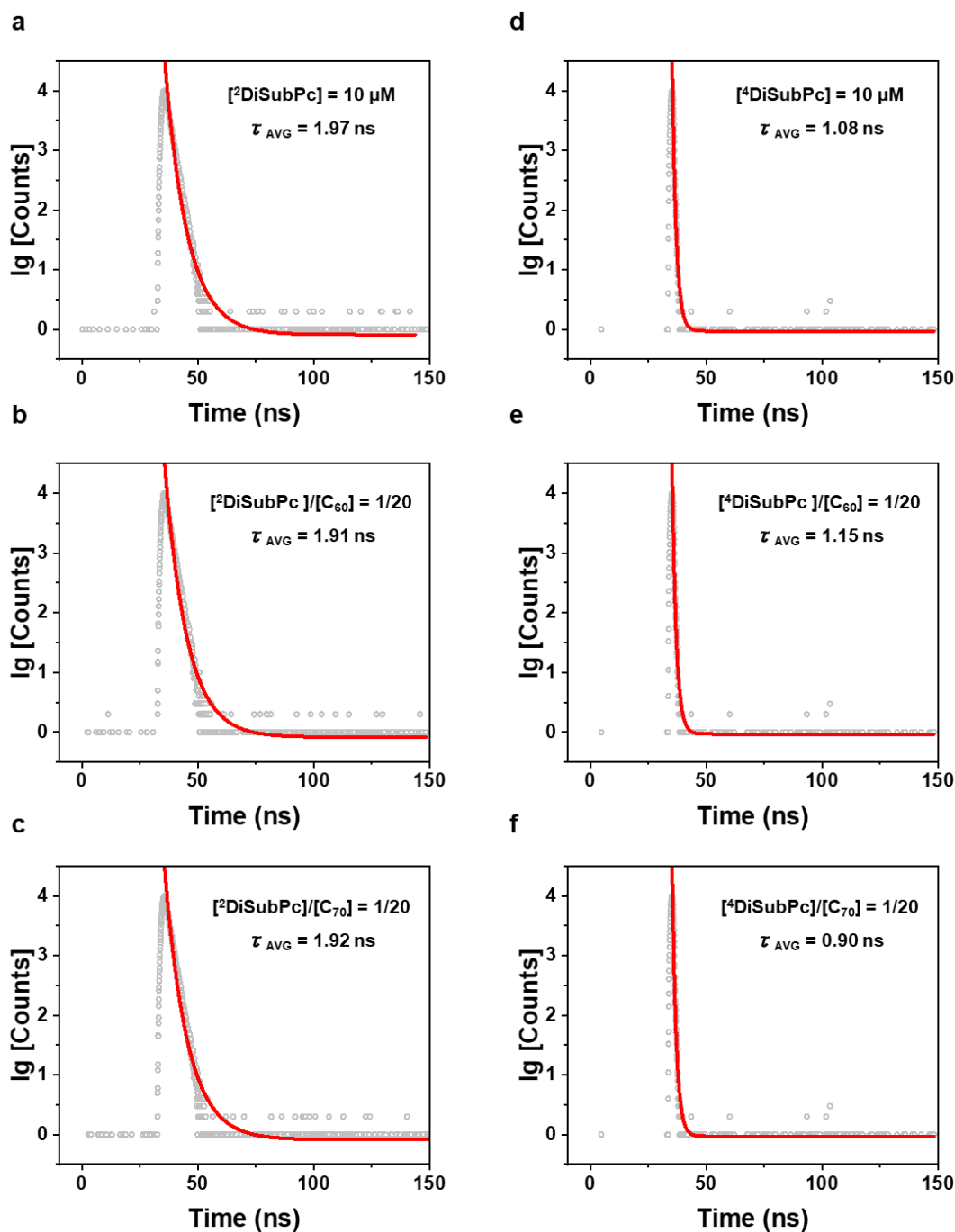


Figure S8. The fluorescence lifetime of $^2\text{DiSubPc}$ ($10.0 \mu\text{M}$ in Toluene) was measured (a) without and with the addition of 20 equivalents of either (b) C_{60} or (c) C_{70} . The fluorescence lifetime of $^4\text{DiSubPc}$ ($10.0 \mu\text{M}$ in Toluene) was measured (d) without and with the addition of 20 equivalents of either (e) C_{60} or (f) C_{70} . The fitted results are shown in red curves.

As showing in figure S6, the addition of fullerenes leads to a gradual quenching of the PL. In general terms, dynamic quenching (quenching agents and fluorophores

collide) and static quenching (quenching agents and fluorophores form the non-fluorescent complex) are the two main causes of fluorescence quenching. The combination quenching effect can be described follows Stern-Volmer equation:

$$F_0/F = (1+K_D[Q]) (1+K_B[Q]) = 1+ (K_D+K_B) [Q]+K_DK_B[Q]^2$$

where F_0 and F are the fluorescent intensity of the fluorophores **DiSubPcs** without and with quenching agent fullerene, K_D is the dynamic quenching constant, $[Q]$ is the concentration of fullerene, K_B is the bimolecular binding constant. Considering the low concentration of $[Q]$, $[Q]^2$ is small enough to be ignored. Since dynamic quenching always leads to changes in fluorescence lifetime, we tested the fluorescence lifetime of **DiSubPcs** solution in toluene before and after adding fullerene. Based on the almost identical fluorescence lifetimes, we judged that not dynamic quenching but static quenching leads to the fluorescence quenching, so the above equation could be estimated as:

$$F_0/F \approx 1+K_B[Q]$$

Therefore, the slope of the above equation, gives the value of the binding constant between **DiSubPc** and C_{60} (C_{70}).

V. Crystal packings

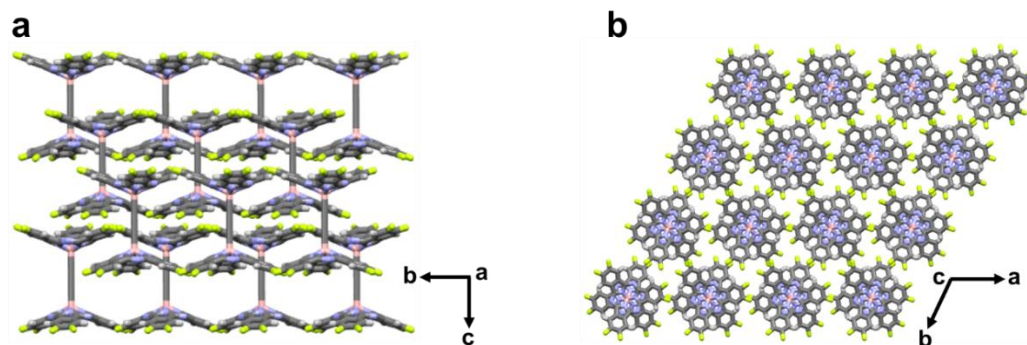


Figure S9. The view from (a) a -axis and (b) c -axis of $^2\text{DiSubPc}$ single crystal.

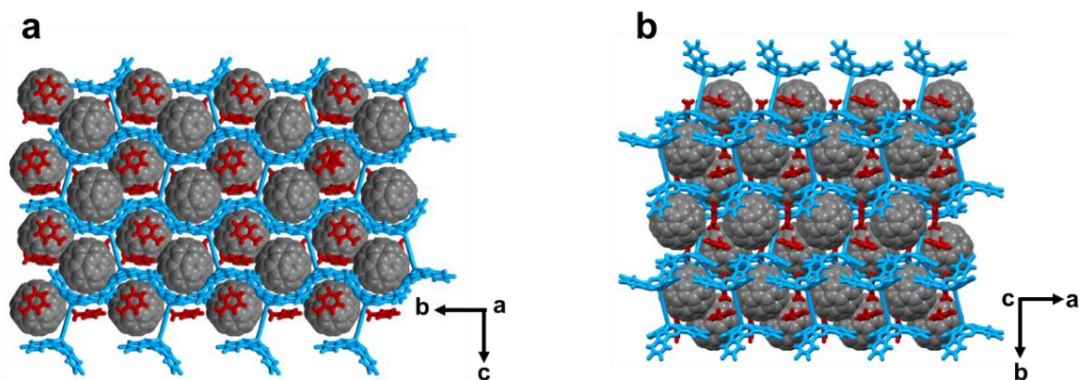


Figure S10. The view from (a) a -axis and (b) c -axis of $^2\text{DiSubPc-C}_{60}$ crystal.

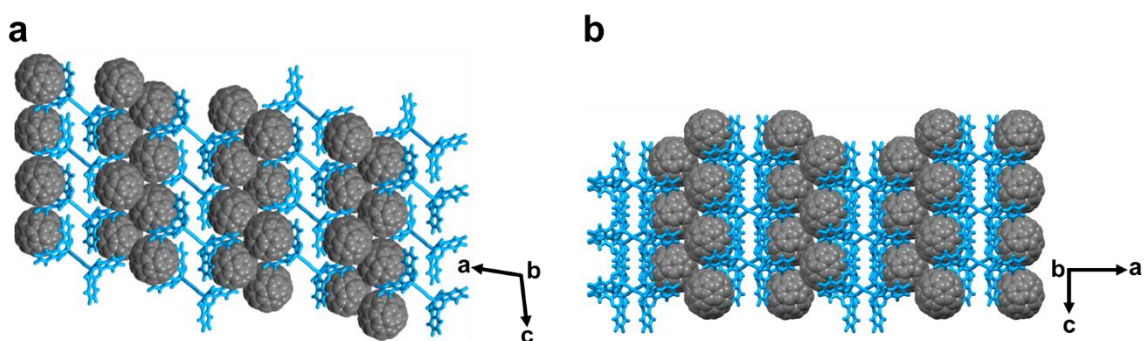
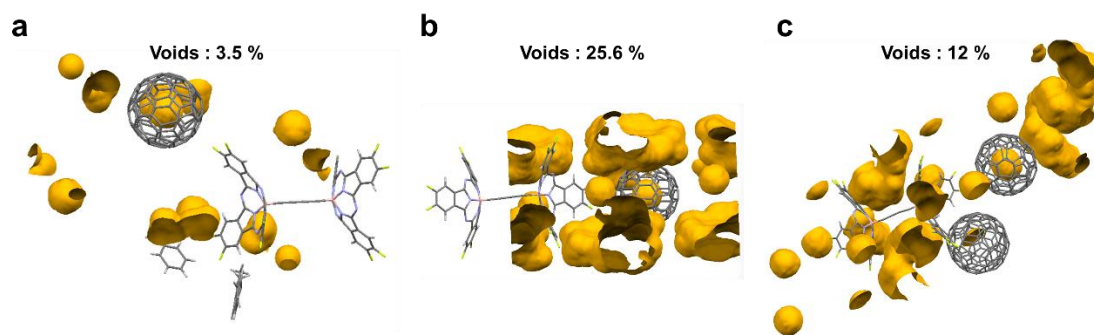


Figure S11. The view from (a) a -axis and (b) b -axis of $^2\text{DiSubPc-C}_{70}$ crystal.

Table S2. Crystallographic data of ²DiSubPc, ²DiSubPc-C₆₀ and ²DiSubPc-C₇₀

Crystals	² DiSubPc	² DiSubPc-C ₆₀	² DiSubPc-C ₇₀
Crystal System	Trigonal	Monoclinic	Monoclinic
Crystal Symmetry	<i>D</i> _{3d}	<i>C</i> _{2h}	<i>C</i> _s
Space Group	<i>R</i> $\bar{3}$ c	<i>P</i> 2 ₁ /n	<i>C</i> c
a [Å]	13.3772(11)	13.238(2)	49.716(8)
b [Å]	13.3772(11)	25.513(3)	11.0347(16)
c [Å]	41.769(6)	17.776(2)	21.540(4)
α [°]	90	90	90
β [°]	90	92.285(5)	103.506(8)
γ [°]	120	90	90
V [Å ³]	6473.14	5998.91	11490.1
R [%]	9.59	13.35	23.67
Rw [%]	6.18	21.71	25.31
B-B Distance [Å]	6.96	6.95	6.90
Fullerene-SubPc Distance [Å]	-	3.15 3.20 3.37	3.03 3.14 3.31
		3.15 3.20 3.37	3.13 3.18 3.26

**Figure S12.** Crystal voids of (a) ²DiSubPc-C₆₀, (b) ²DiSubPc-C₆₀ without toluene, and (c) ²DiSubPc-C₇₀.

Crystal void analysis indicated that the void volumes in ²DiSubPc-C₆₀ and ²DiSubPc-C₇₀ cocrystals accounted for 3.5% and 12% of the unit cell volume, respectively. However, upon removal of toluene from the ²DiSubPc-C₆₀ structure, the voids increased significantly from 3.5% to 25.6%, suggesting that the resulting voids are too large to sustain the original packing arrangement, likely leading to structural collapse. At the same time, the voids in ²DiSubPc-C₇₀ cocrystals is insufficient to accommodate toluene molecules.

VI. Theoretical calculations

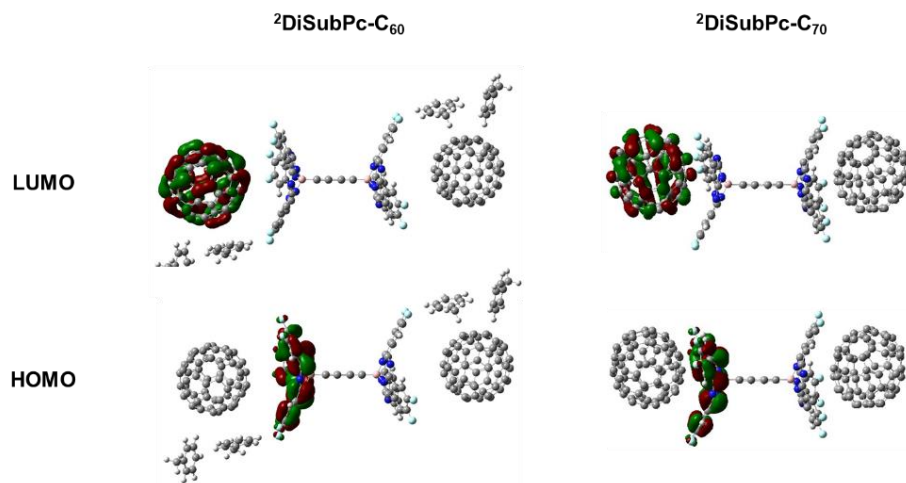


Figure S13. Calculated FMO distributions of ${}^2\text{DiSubPc-C}_{60}$ and ${}^2\text{DiSubPc-C}_{70}$.

VII. Photophysical properties

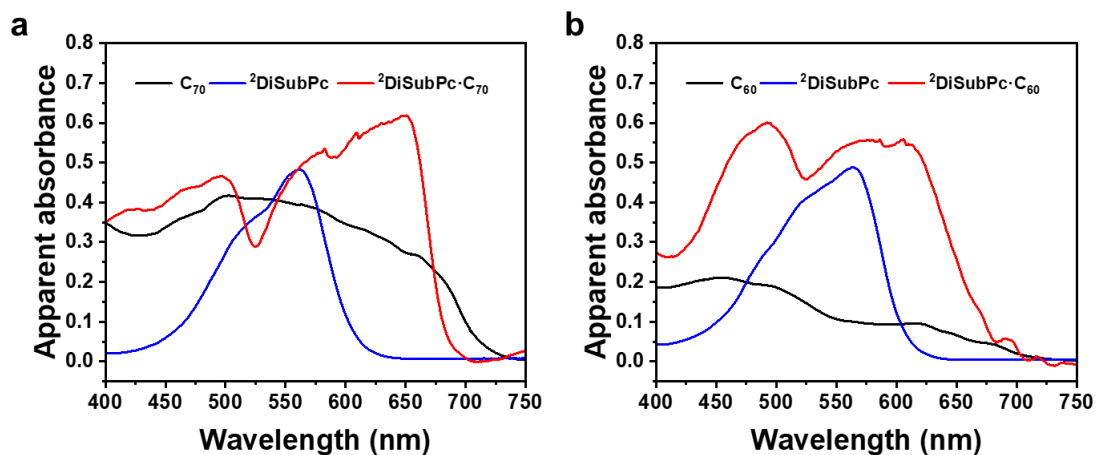


Figure S14. Apparent absorption spectra. (a) Comparison of UV-vis-NIR absorption of C_{70} and $^2\text{DiSubPc}$ single crystals with $^2\text{DiSubPc}\cdot C_{70}$ cocrystals. (b) Comparison of UV-vis-NIR absorption of C_{60} , and $^2\text{DiSubPc}$ single crystals with $^2\text{DiSubPc}\cdot C_{60}$ cocrystals.

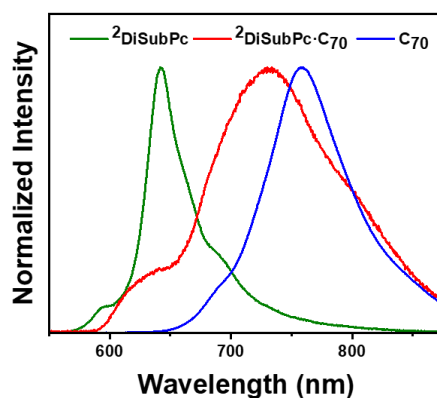


Figure S15. PL Comparison of $^2\text{DiSubPc}$, $^2\text{DiSubPc}\cdot C_{70}$ and C_{70} crystals at 300 K.

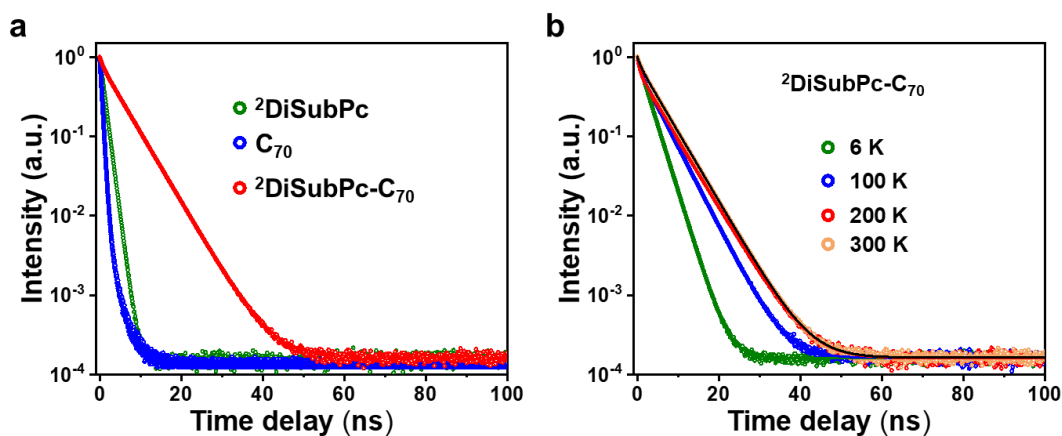


Figure S16. (a) PL lifetime Comparison of $^2\text{DiSubPc}$, C_{70} , and $^2\text{DiSubPc}\cdot C_{70}$ crystals at 300 K. (b) Temperature-dependent PL lifetime spectra of $^2\text{DiSubPc}\cdot C_{70}$ cocrystals.

Table S3. Temperature-dependent PL lifetime of ²DiSubPc-C₇₀ cocrystals.

² DiSubPc-C ₇₀	6 K	100 K	200 K	300 K
A ₁	54.8%	39.9%	19.3%	15.5%
τ ₁ (ns)	0.73	0.71	0.71	0.73
A ₂	45.2%	60.1%	80.7 %	84.5%
τ ₂ (ns)	3.72	4.20	4.62	4.96

Table S4. The PL lifetime of ²DiSubPcs, ²DiSubPcs-C₆₀, ²DiSubPcs-C₇₀ and C₇₀ at 300 K.

Samples	² DiSubPc	² DiSubPc-C ₆₀	² DiSubPc-C ₇₀	C ₇₀
A ₁	100%	100%	15.5%	100%
τ ₁ (ns)	1.12	0.89	0.73	0.86
A ₂	-	-	84.5%	-
τ ₂ (ns)	-	-	4.96	-

Temperature-dependent PL spectra

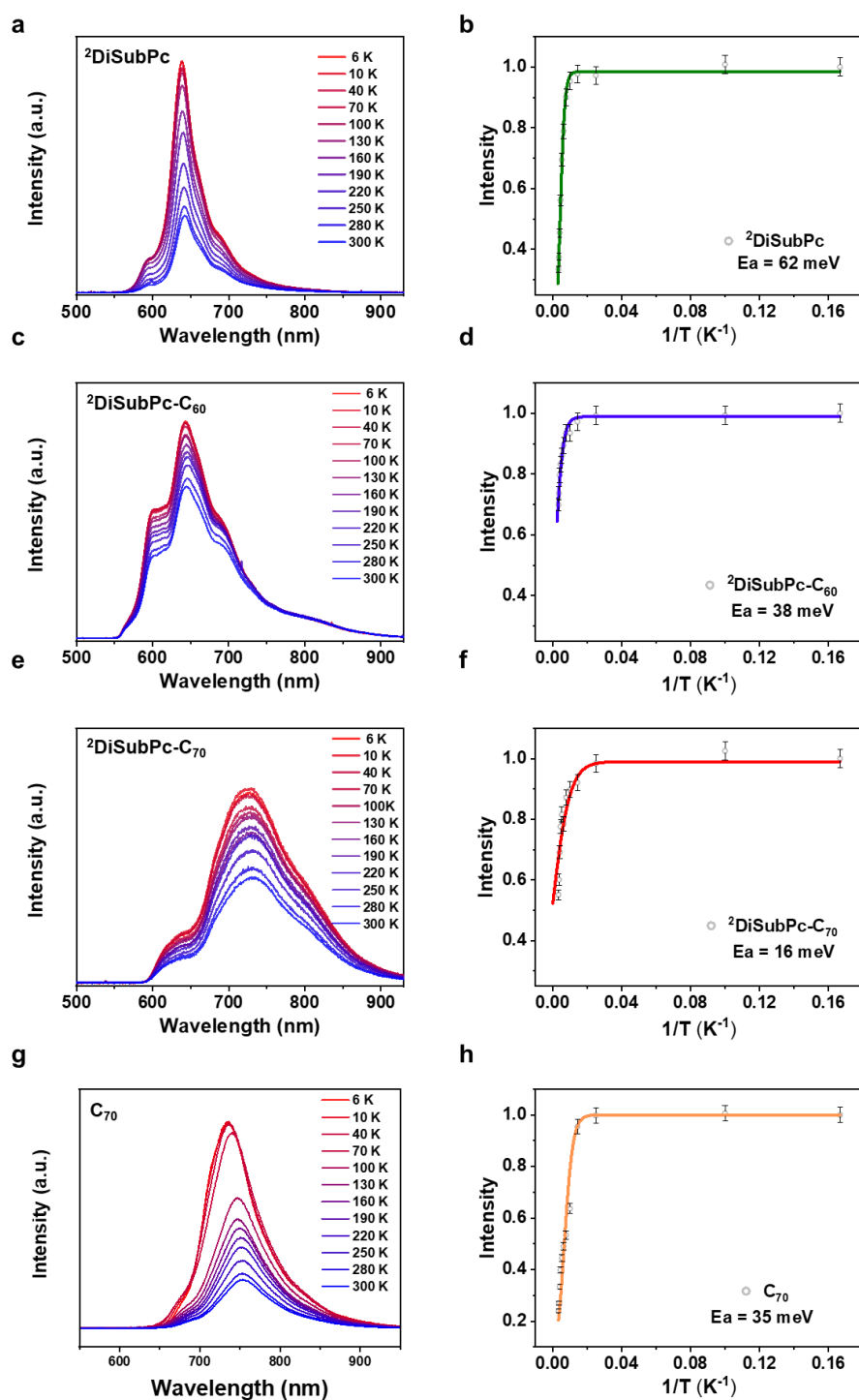


Figure S17. (a, c, e, g) The temperature-dependent PL spectra from 6 K to 300 K; (b, d, f, h) The corresponding integral of fluorescence intensity (black dots) employed for linear fitting, with the fitting curve and obtained activation energy displayed. Note: measurements were repeated three times at each temperature to mitigate systematic errors arising from fluctuations in light intensity, thermal drift, and sample positioning. Error bars represent the mean values $\pm$ the maximum offset across replicate measurements.

$$I(T) = \frac{I_0}{1 + A \cdot e^{-E_a/(k_B T)}}$$

In the equation, $I(T)$ and I_0 refer to photoluminescence intensity of sample in specific temperature and low temperature, A is the ratio factor, E_a is the activation energy of nonradiative process, k_B is the Boltzmann constant and T is the specific temperature.

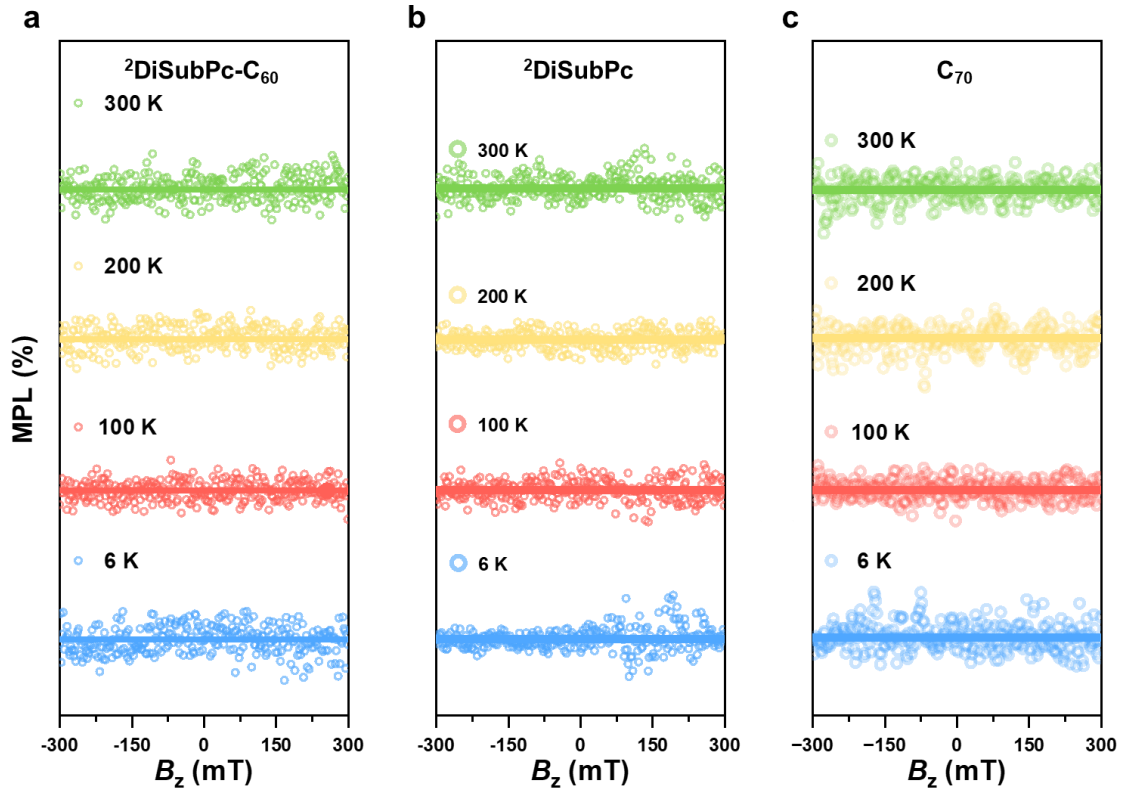


Figure S18. The magneto-photoluminescence spectra at different temperatures of (a) ${}^2\text{DiSubPc-C}_{60}$, (b) ${}^2\text{DiSubPc}$ and (c) C_{70} .

VIII. Photothermal conversion

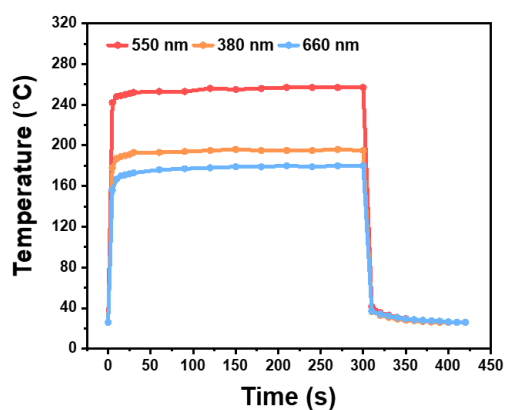


Figure S19. Wavelength-dependent photothermal conversion experiments of $^2\text{DiSubPc-C}_{70}$.

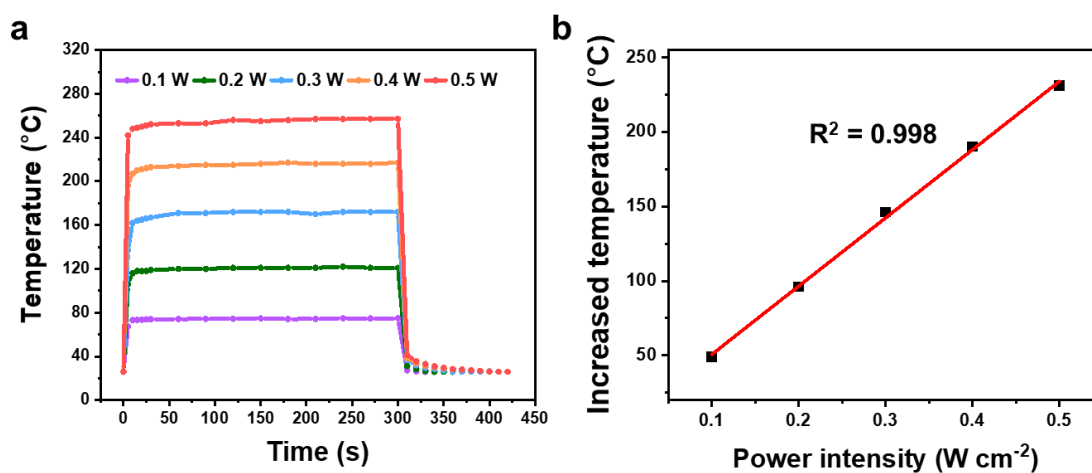


Figure S20. (a) Intensity-dependent photothermal conversion experiments and (b) linear relationship between light intensity and increased temperature of $^2\text{DiSubPc-C}_{70}$.

Table S5. The photothermal properties of **²DiSubPcs-C₇₀** compared with previous organic solids in the literature.

Samples	Increased temperature (°C)/light intensity (W cm ⁻²)	Normalized increased temperature (°C) in 0.5 W cm ⁻² [a]	References
²DiSubPc-C₇₀	213/0.5	213	This work
TCNQ-Tp-Py	130/0.9	72	<i>Angew. Chem. Int. Ed.</i> 2025 , 64, e202413805
CF cocrystals	60/0.276	109	<i>ACS. Nano.</i> 2022 , 16, 15000
TAHC	60/0.5	60	<i>Angew. Chem. Int. Ed.</i> 2022 , 134, e202202571
TPP-2IND	160/0.9	89	<i>Aggregate</i> 2024 , 5, e535
DDPA-PDN	114/0.8	71	<i>Adv. Funct. Mater.</i> 2021 , 31, 2106247
FCDTDPP	100/0.5	100	<i>Angew. Chem. Int. Ed.</i> 2025 , 64, e202418047
GDPA-QCN	30/0.1	150	<i>Angew. Chem. Int. Ed.</i> 2022 , 61, e202117087
COF-Azu-TP	200/1	100	<i>Small</i> 2024 , 20, 2307635
P(DPP) ₄	115/0.8	72	<i>Dyes and Pigments</i> 2021 , 192, 109460
PPBBT	48/0.8	30	<i>Nat. Commun.</i> 2019 , 10, 1192
CR-TPE-T	100/0.8	63	<i>Adv. Mater.</i> 2022 , 34, 2108048
CR-DPA-T	110/0.8	69	<i>Adv. Mater.</i> 2020 , 32, 1908537

[a] Assuming a linear correlation between the added temperature and light intensity, we give the value of the normalized temperature raise for these materials irradiated with the light of 0.5 W cm⁻².

IX. References

(1).(a) Afraj, S. N., Jiang, B. H., Su, Y. W., Yang, C. H., Shih, H. S., Velusamy, A., Ni, J. S., Ezhumalai, Y., Su, T. Y., Liu, C. L., Yau, S., Chen, C. P., Chen, M. C., Dicyclopentadithienothiophene-based non-fullerene acceptors for ternary blend organic photovoltaics. *J. Mater. Chem. C*. **2024**, *12*, 2247-2257; (b) Bukuroshi, E., Vestfrid, J., Gross, Z., Bender, T. P., Fluorinated boron subphthalocyanines: Lewis acid based templating chemistry facilitates random halide exchange, and fluoride *versus* chloride affects the basic photophysical properties and the solid-state arrangement. *New. J. Chem.* **2019**, *43*, 16730-16737; (c) Bukuroshi, E., Petersen, A. U., Brolos, L., Bender, T. P., Nielsen, M. B., Exploring the synthesis and electronic properties of axially substituted boron subphthalocyanines with carbon-based functional groups. *Eur. J. Inorg. Chem.* **2020**, *2020*, 3481-3495.

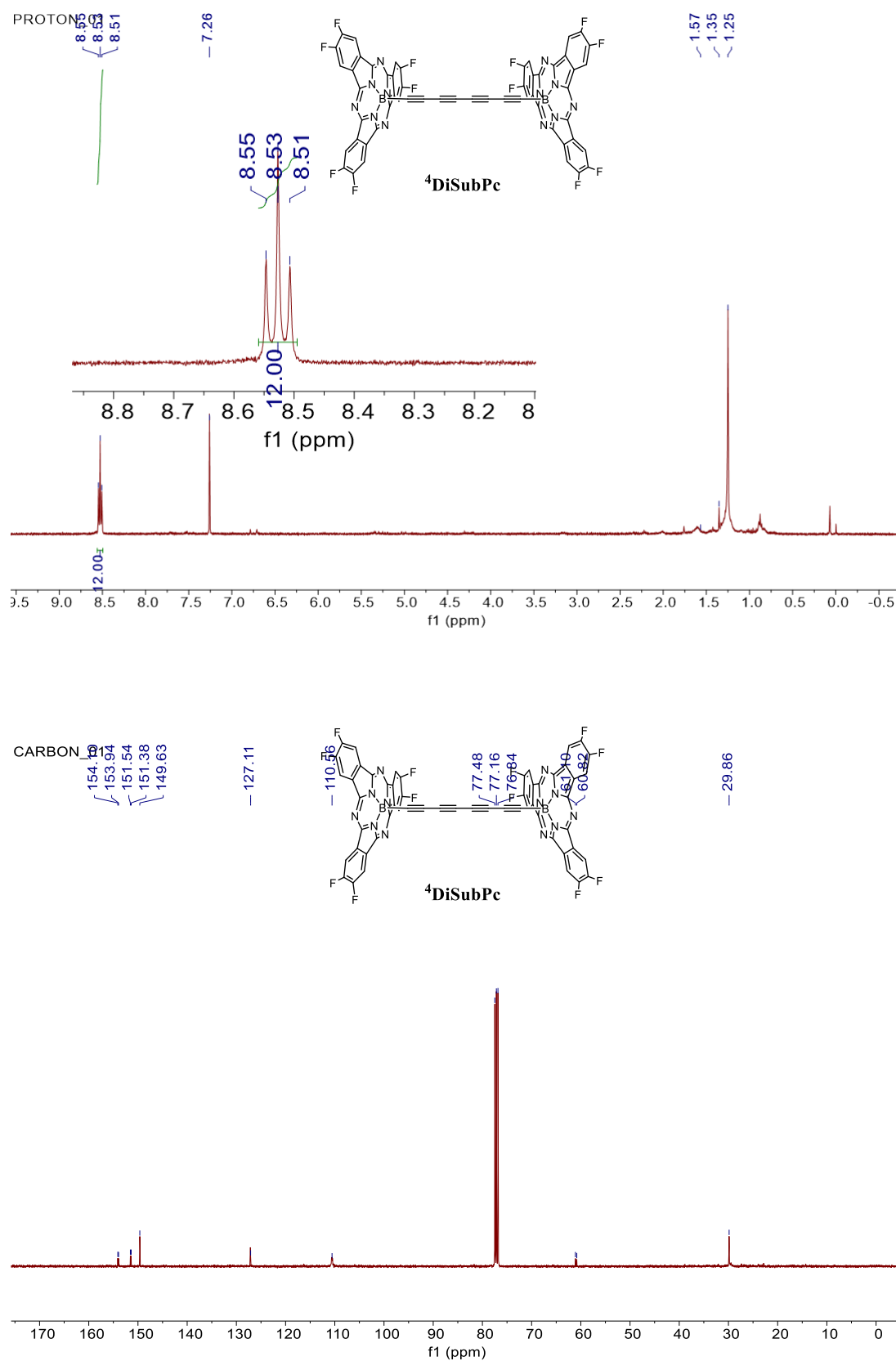


Figure S22. ¹H-NMR and ¹³C-NMR spectra of **⁴DiSubPc**